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**SYSTEM MODELLING AND DESIGN: ROAD SIGN AND ROAD STATE MOBILE
NOTIFICATION APPLICATION**

GROUP 5

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1. Introduction

Effective system modeling and design is a critical step in developing complex software applications. It involves creating visual representations and abstractions of the system's structure, behavior, and interactions to better understand and communicate the system requirements and architecture. This report presents the system modeling and design for a Road Sign and Road State Mobile Notification Application using Unified Modeling Language (UML) diagrams.

The key objectives of this system modeling exercise are to:

- Capture functional requirements through use case diagrams and interaction models
- Model the system's static structure using class diagrams and deployment diagrams
- Capture the dynamic behavior with sequence diagrams and dataflow diagrams
- Define the system context and external interactions through context diagrams
- Provide a comprehensive visual representation of the system architecture

1.1 Purpose

The purpose of this document is to design and model the requirements for a road sign, road state, and notification app that provides users with real-time traffic updates, location-based services, and other advanced features. The system aims to enhance road safety and improve navigation efficiency through intelligent information delivery.

1.2 Scope

The scope of this system modeling includes the development of UML diagrams that represent:

- System boundaries and external entity interactions
- Data flow and processing within the system
- User interactions and system use cases
- Object-oriented design structure and relationships
- Temporal behavior and message sequences

- Physical deployment architecture

2. System Design Overview

The Road Sign and Road State Mobile Notification Application follows a client-server architecture with real-time data integration capabilities. The system serves multiple user types including drivers, pedestrians, and transport agencies, providing them with critical road information, navigation assistance, and safety alerts.

2.1 System Architecture

The system architecture is based on a client-server model with the following key components:

- **Mobile Application:** User-facing client providing core functionalities and user interface
- **Backend Server:** Server-side component responsible for data processing and management
- **Data Sources:** External data providers supplying traffic, weather, and road information
- **Database System:** Persistent storage for road signs, user data, and system information
- **Communication Layer:** Protocols enabling data exchange between system components

2.2 Key Features

The system provides the following essential features:

- Real-time traffic updates and road condition monitoring
- Location-based services with GPS integration
- Road sign recognition and display capabilities
- User feedback and crowd-sourced reporting
- Audio notifications and customizable alerts
- Offline mode functionality for areas with poor connectivity

- Navigation assistance with route optimization

3. UML Diagrams

3.1 Context Diagram

The context diagram provides a high-level view of the system showing its interactions with external entities and the flow of information across system boundaries.

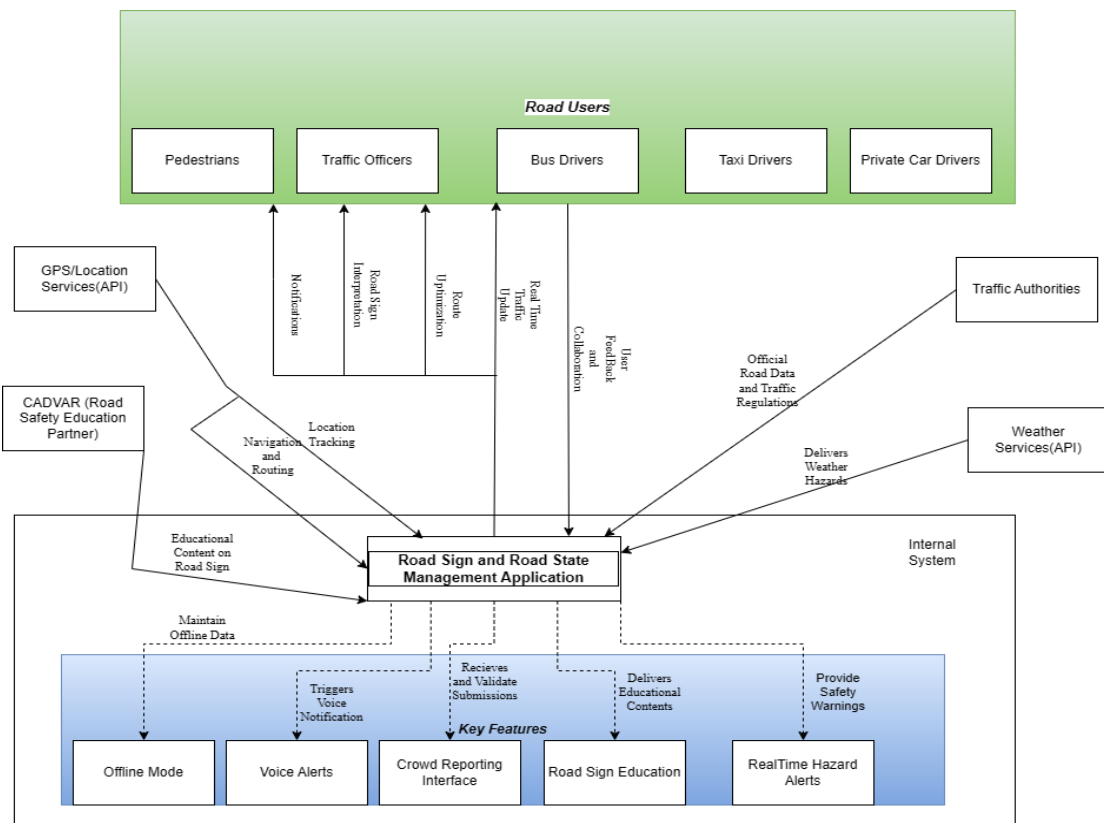
Purpose: To illustrate the system's position within its environment and identify all external entities that interact with the system.

Key External Entities:

- **Drivers:** Primary users receiving road updates and navigation assistance
- **Pedestrians/Transport Agencies:** Secondary users accessing road information
- **GPS System:** Provides real-time location data for context-aware services
- **Data Providers:** Supply real-time traffic, weather, and road condition data
- **Admin/Programmers:** System administrators managing configuration and maintenance

Data Flows:

- Input preferences and location data from users
- Real-time road conditions and notifications to users
- Location data from GPS systems
- Traffic and weather data from external providers
- System configuration and maintenance data from administrators



3.2 Dataflow Diagram

The dataflow diagram illustrates how data moves through the system, showing the processes that transform data and the data stores that hold information.

Purpose: To model the functional requirements of the system by showing data transformation processes and information flow.

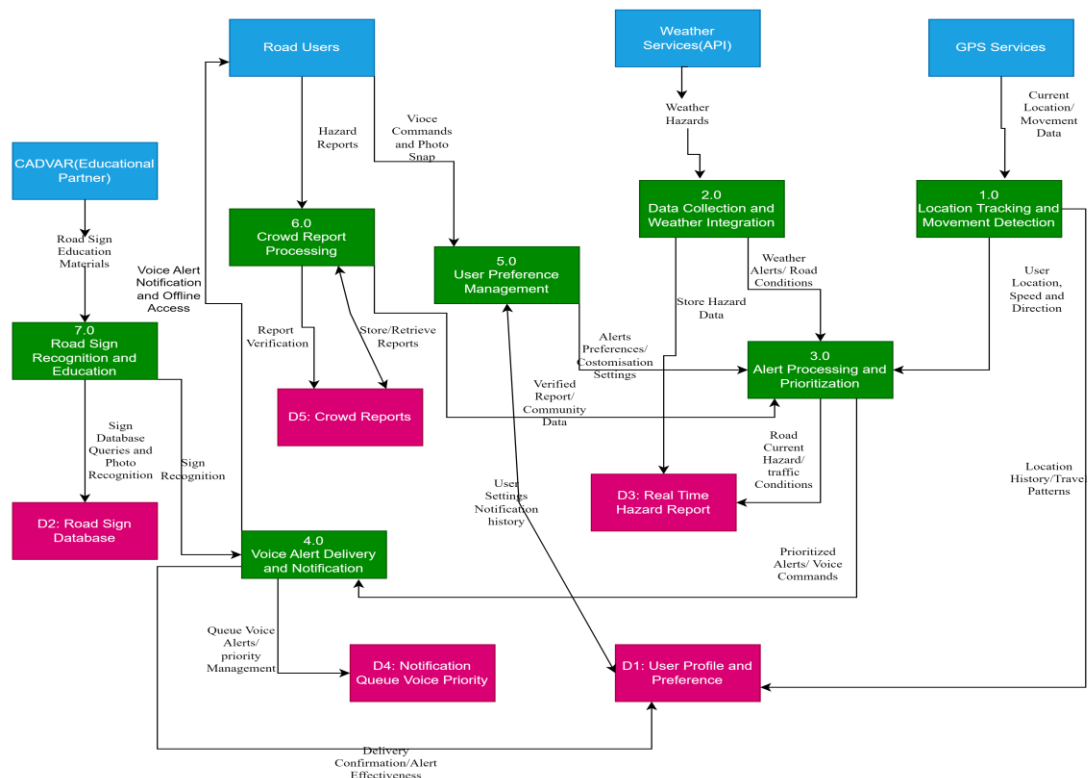
Key Processes:

- **Data Collection and Weather Monitoring (2.0):** Gathers real-time traffic and weather information
- **Location Tracking and Movement Detection (1.0):** Processes GPS data and user movement
- **Alert Processing and Prioritization (3.0):** Manages alert generation and delivery
- **User Preference Management (5.0):** Handles user settings and customization

- **Crowd Report Processing:** Manages user-generated reports and feedback

Data Stores:

- **Road Sign Database (D2):** Stores comprehensive road sign information
- **Real Time Hazard Report (D4):** Contains current hazard and incident data
- **User Profile and Preferences (D1):** Maintains user settings and preferences
- **Notification Queue Voice Priority (D3):** Manages alert delivery queue



3.3 Use Case Diagram

The use case diagram represents the functional requirements of the system by showing the interactions between actors and the system's use cases.

Purpose: To capture the functional requirements from a user perspective and show the relationships between different system functionalities.

Primary Actors:

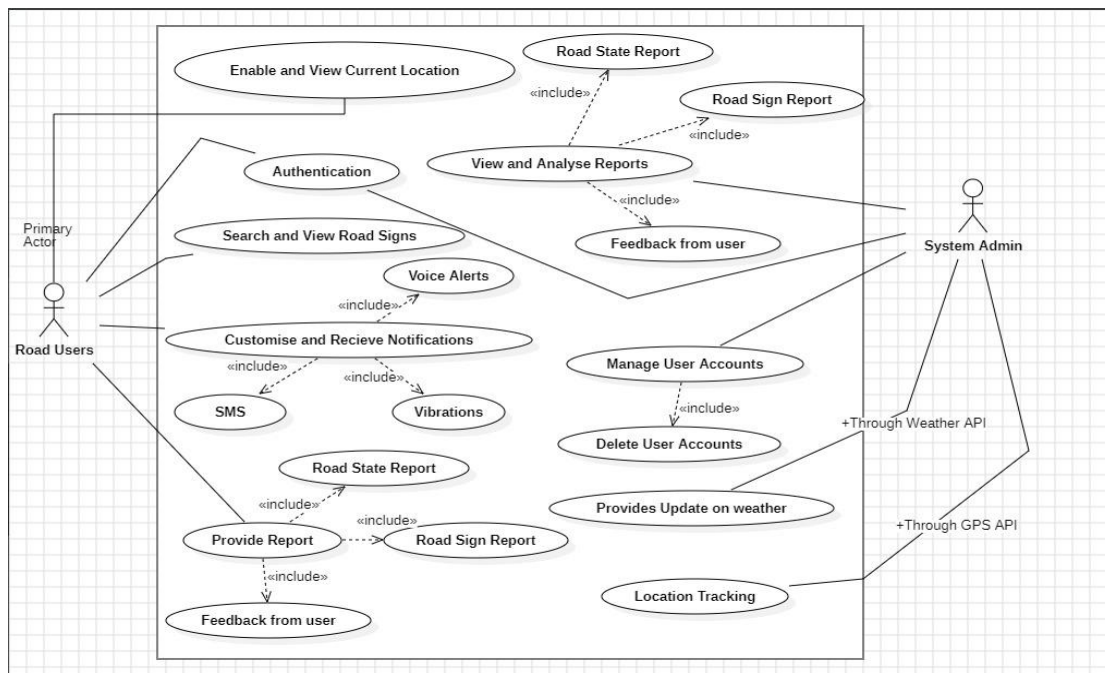
- **Road Users:** Including drivers, pedestrians, and transport personnel
- **System Admin:** Administrative personnel managing the system

Key Use Cases:

- **Authentication:** User login and registration processes
- **Enable and View Current Location:** Location services activation and display
- **Search and View Road Signs:** Road sign directory access and search functionality
- **Voice Alerts:** Audio notification management and delivery
- **Customise and Receive Notifications:** Notification preference settings (SMS, Vibrations)
- **View and Analyse Reports:** Report generation and analysis capabilities
- **Feedback from User:** User feedback submission and processing
- **Manage User Accounts:** Administrative user management functions
- **Location Tracking:** GPS-based location monitoring and services
- **Provides Update on Weather:** Weather information integration through API

Use Case Relationships:

- **Include Relationships:** Show mandatory sub-functionality inclusion
- **Extend Relationships:** Represent optional feature extensions
- **Generalization:** Demonstrate inheritance between related use cases



3.4 Sequence Diagrams

Sequence diagrams show the temporal order of interactions between objects for specific scenarios, illustrating how processes operate with one another and in what order.

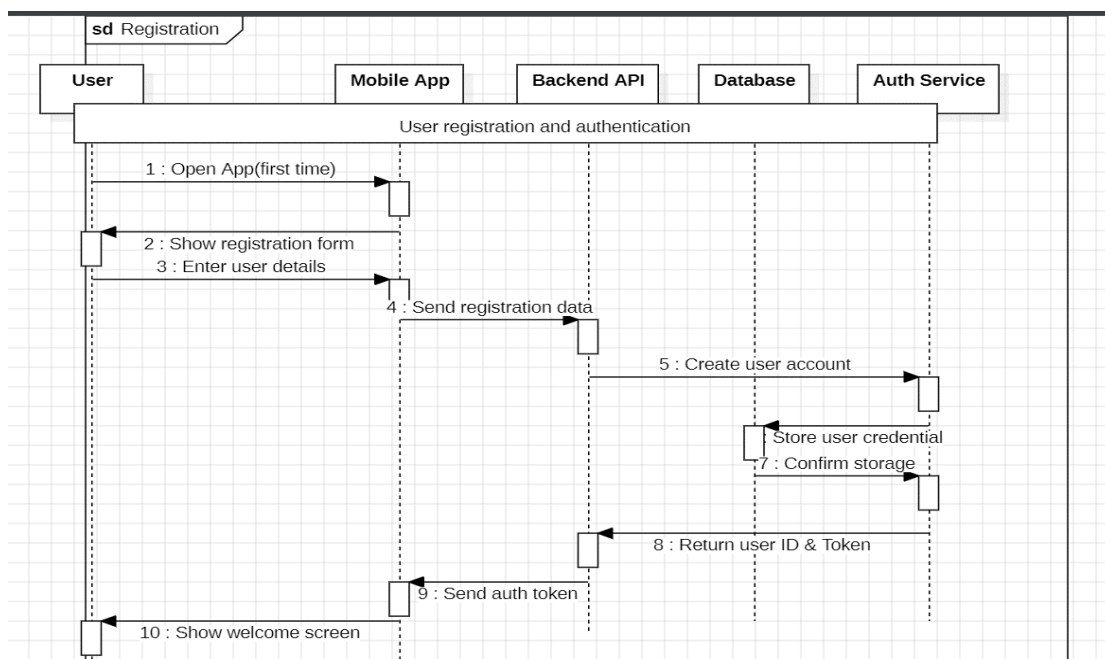
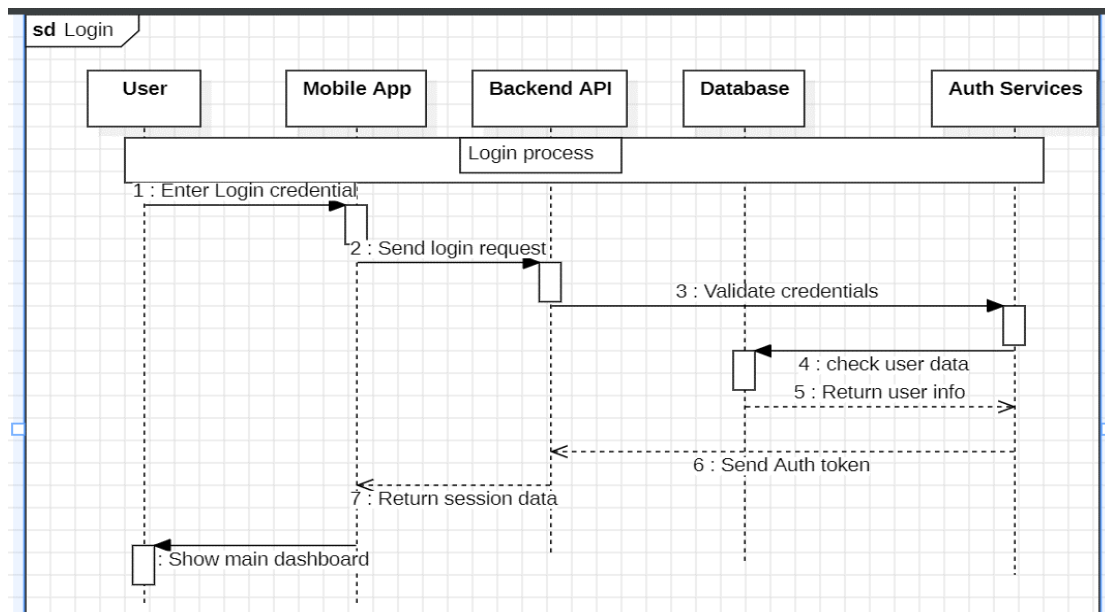
Purpose: To model the dynamic behavior of the system by showing object interactions over time for specific use cases.

3.4.1 Login/Register Sequence Diagram

Participants: User, Mobile Application, Authentication Server, User Database

Interaction Flow:

1. User initiates login/registration request
2. Mobile application validates input data
3. Authentication server processes credentials
4. Database verification and user record management
5. Success/failure response returned to user



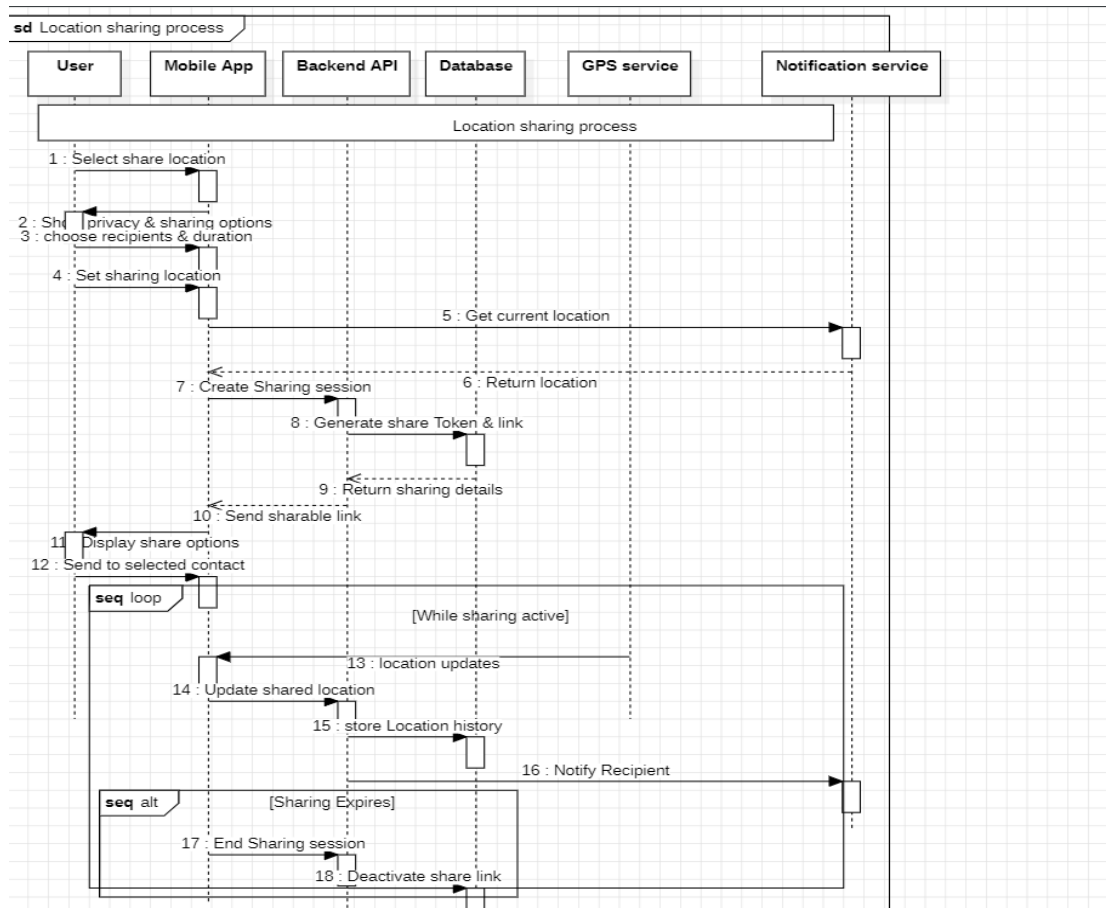
3.4.2 Location Sharing Sequence Diagram

Participants: User, Mobile Application, GPS Service, Location Manager, Contact Database

Interaction Flow:

1. User activates location sharing feature
2. GPS service provides current location coordinates

3. Location manager processes and validates location data
4. Contact selection and permission verification
5. Location data transmission to selected contacts

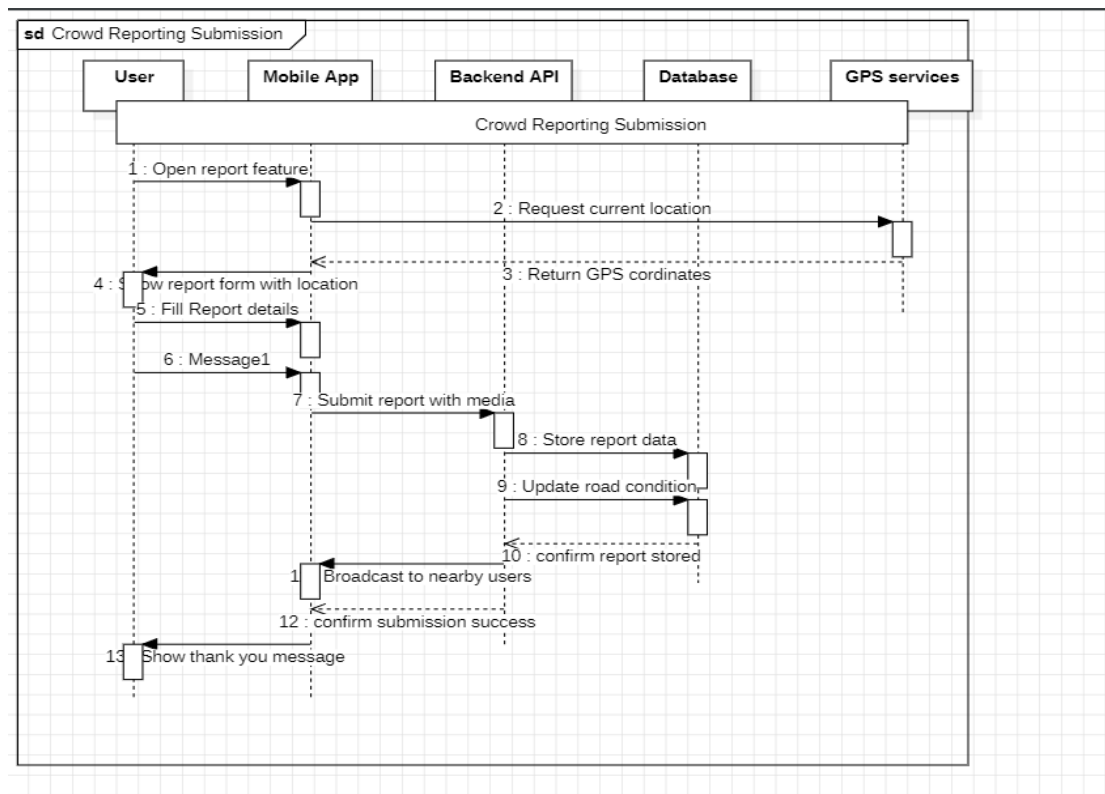


3.4.3 Crowd Reporting Sequence Diagram

Participants: User, Mobile Application, Report Processing Server, Verification System, Report Database

Interaction Flow:

1. User submits crowd-sourced report
2. Mobile application captures report data and location
3. Report processing server validates and categorizes report
4. Verification system confirms report accuracy
5. Report integration into system alerts and database storage

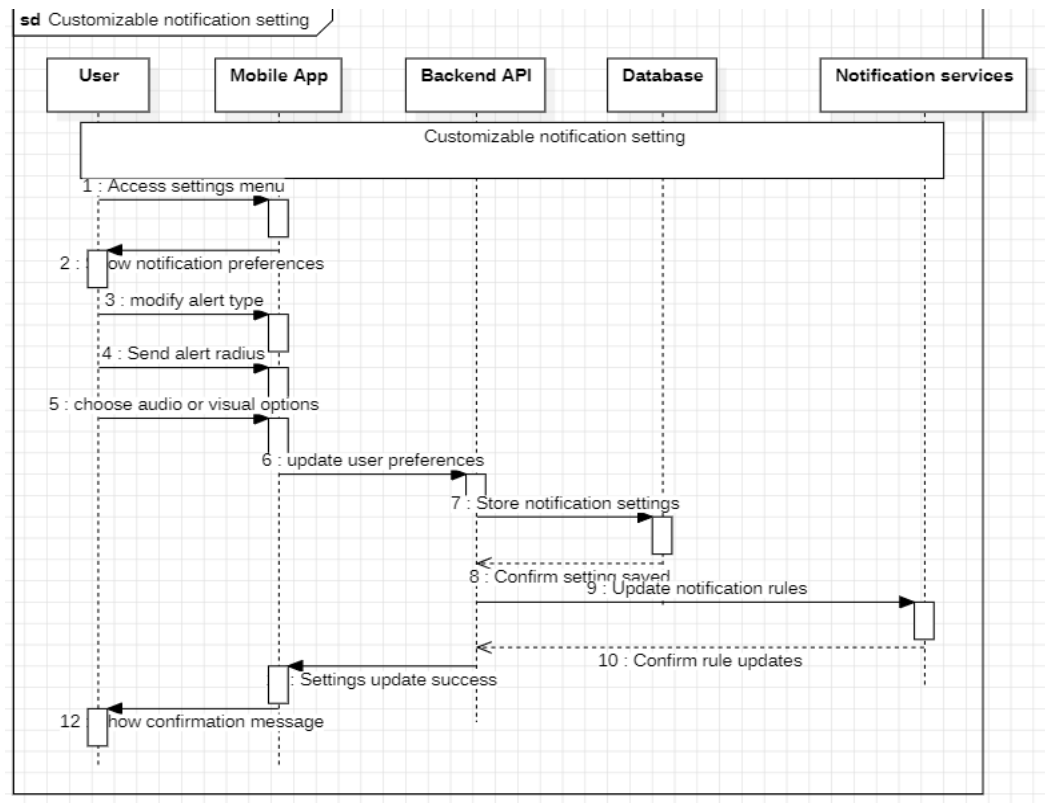


3.4.4 Customizable Notification Sequence Diagram

Participants: User, Mobile Application, Notification Manager, Preference Database, Alert System

Interaction Flow:

1. User accesses notification settings interface
2. Preference selection and customization options
3. Settings validation and database storage
4. Notification manager configuration update
5. Confirmation and system activation

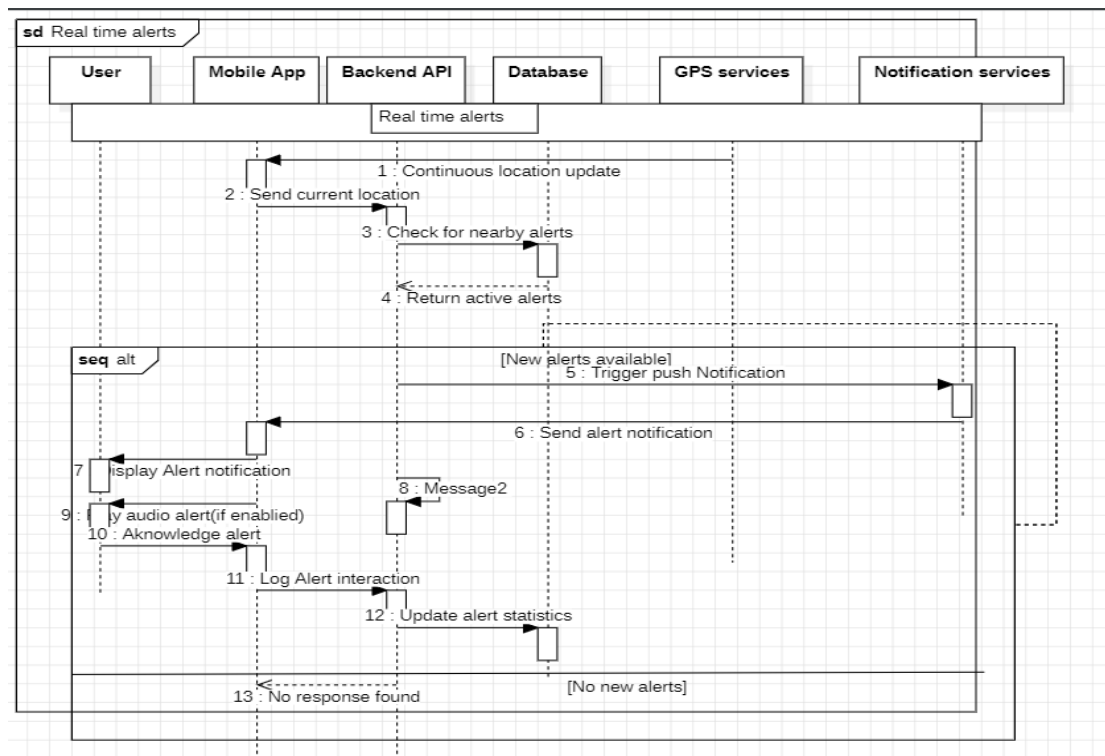


3.4.5 Real-Time Alert Sequence Diagram

Participants: Data Provider, Alert System, Processing Engine, Notification Manager, Mobile Application, User

Interaction Flow:

1. Data provider sends real-time traffic/road condition data
2. Alert system processes incoming data for relevant alerts
3. Processing engine determines affected users and alert priority
4. Notification manager delivers alerts based on user preferences
5. Mobile application displays alerts to users

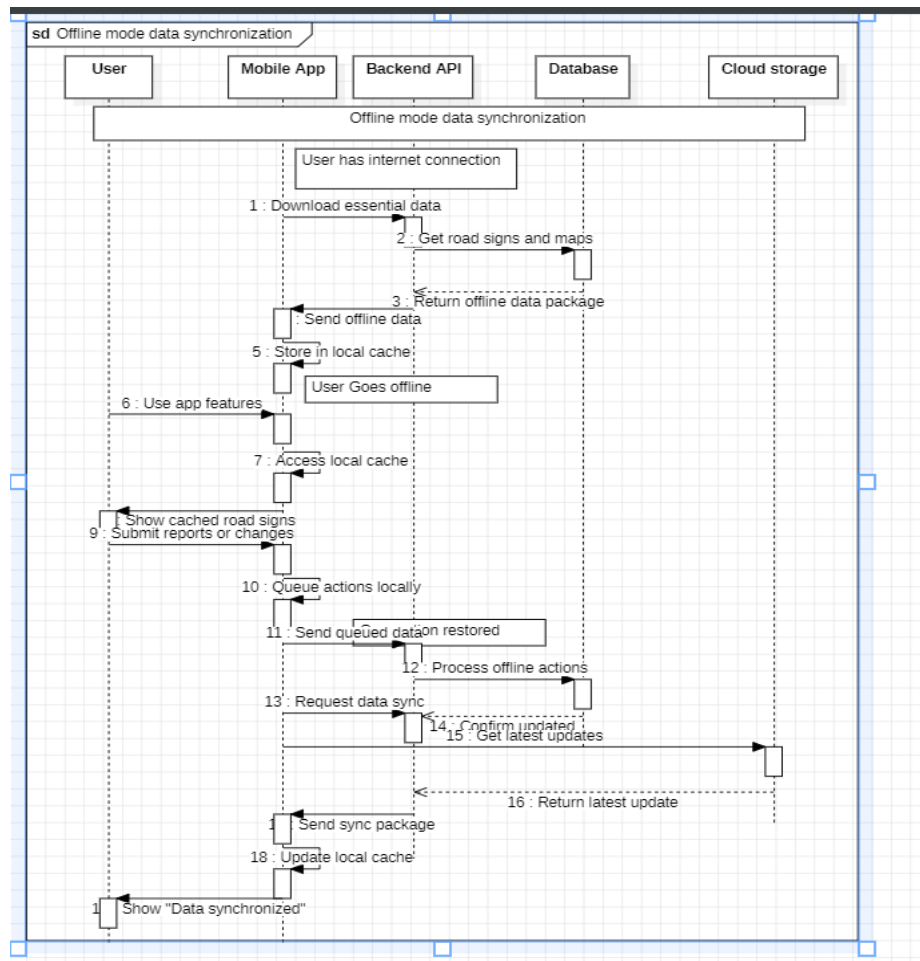


3.4.6 Offline Mode Sequence Diagram

Participants: Mobile Application, Network Monitor, Cache Manager, Local Storage, Sync Service

Interaction Flow:

1. Network monitor detects connectivity loss
2. Mobile application switches to offline mode
3. Cache manager retrieves locally stored data
4. Limited functionality provided using cached information
5. Automatic synchronization when connectivity is restored

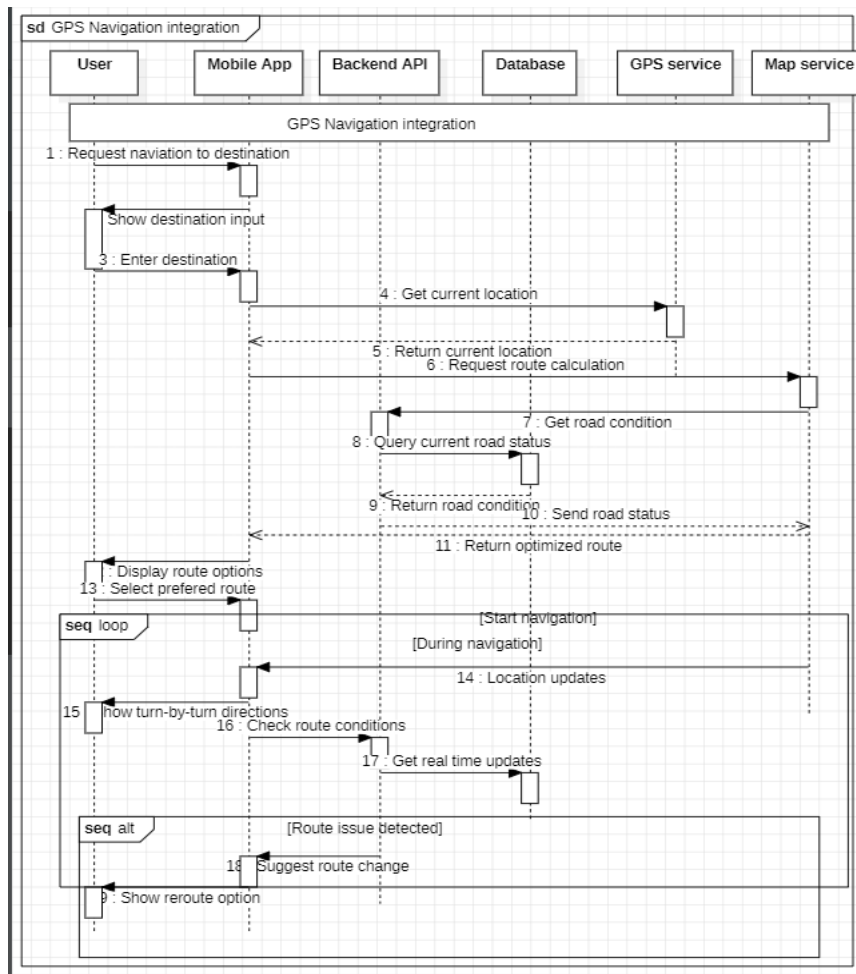


3.4.7 GPS Navigation Sequence Diagram

Participants: User, Navigation Interface, GPS Service, Route Calculator, Map Display, Voice System

Interaction Flow:

1. User inputs destination through navigation interface
2. GPS service provides current location coordinates
3. Route calculator determines optimal path using traffic data
4. Map display renders navigation interface with route
5. Voice system provides turn-by-turn audio guidance



3.5 Class Diagram

The class diagram shows the static structure of the system by displaying the system's classes, their attributes, operations, and the relationships among objects.

Purpose: To represent the object-oriented design of the system showing classes, their properties, methods, and relationships.

Key Classes:

3.5.1 Core Classes

- **User:** Manages user identification, authentication, and profile information
 - Attributes: userId, username, email
 - Methods: register(), login(), updateProfile()

- **Location:** Handles GPS coordinates and location-based operations
 - Attributes: latitude, longitude, timestamp
 - Methods: getLocation(), updateLocation()
- **UserPreferences:** Stores user customization settings
 - Attributes: notifications, voiceEnabled
 - Methods: update(), getPrefs()

3.5.2 Alert Management Classes

- **RealTimeAlert:** Base class for all alert types
 - Attributes: alertId, alertType, message
 - Methods: create(), broadcast()
- **TrafficAlert:** Specialized alert for traffic conditions
 - Attributes: trafficLevel, estimatedDelay
 - Methods: getStatus(), suggest()
- **AccidentAlert:** Handles accident and emergency notifications
 - Attributes: accidentType, vehiclesInvolved
 - Methods: notificationEmergency()

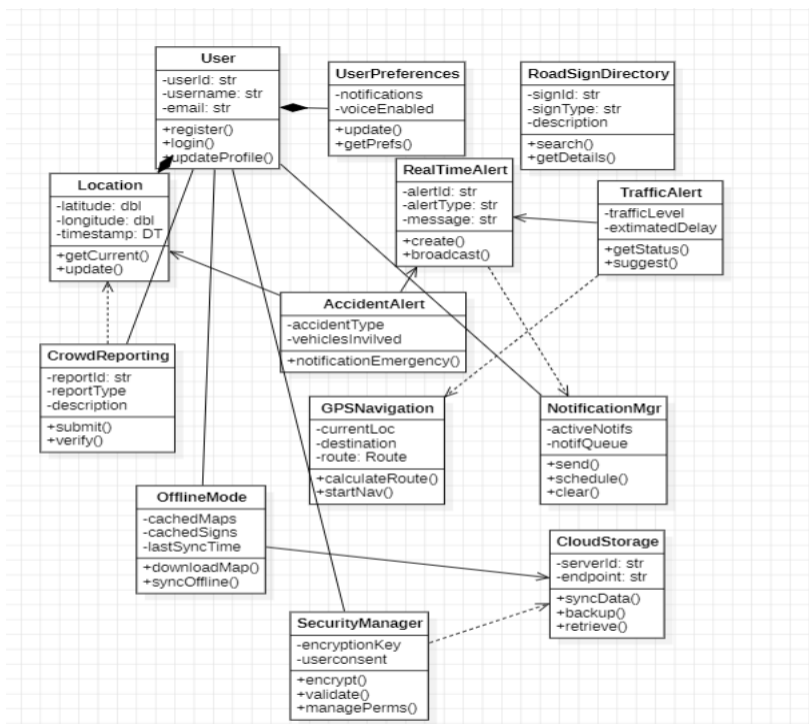
3.5.3 System Management Classes

- **NotificationManager:** Controls alert delivery and queue management
 - Attributes: activeNotifs, notifQueue
 - Methods: send(), schedule(), clear()
- **GPSNavigation:** Provides navigation services and route calculation
 - Attributes: currentLoc, destination, route
 - Methods: calculateRoute(), startNav()
- **SecurityManager:** Handles system security and data protection
 - Attributes: encryptionKey, userConsent
 - Methods: encrypt(), validate(), managePerms()

Class Relationships:

- **Association:** Direct relationships between classes
- **Inheritance:** Specialized classes inheriting from base classes

- **Composition:** Strong ownership relationships
- **Aggregation:** Weak ownership relationships



3.6 Deployment Diagram

The deployment diagram shows the physical deployment of software components on hardware nodes, illustrating the system's runtime architecture.

Purpose: To represent the physical architecture of the system showing how software components are distributed across hardware infrastructure.

Hardware Nodes:

3.6.1 Client-Side Deployment

- **Android Device:** Mobile smartphones and tablets
 - Components: RoadSmart App, Offline Cache
 - Protocols: HTTPS, API Calls
- **Cloud Services:** External service integration
 - Components: Firebase Storage, Google Maps API, OpenWeather API

- Connections: API Calls, Firebase SDK

3.6.2 Server-Side Deployment

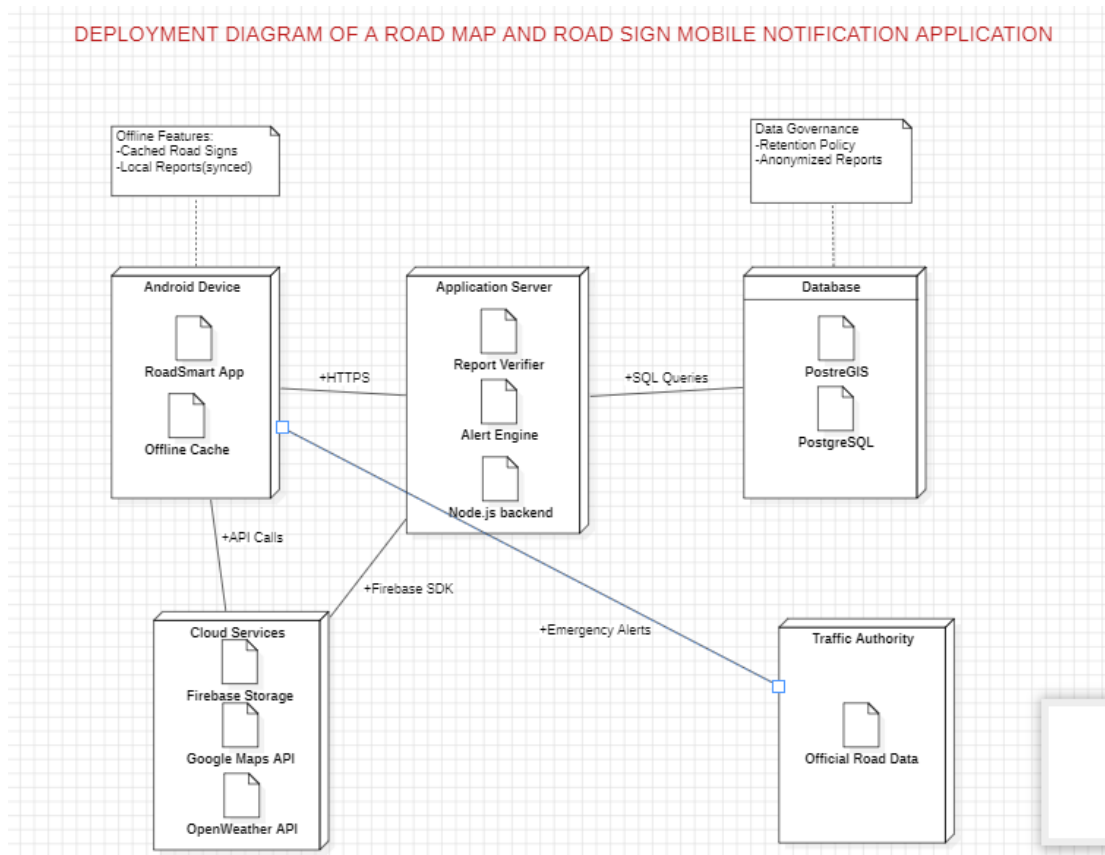
- **Application Server:** Core business logic hosting
 - Components: Report Verifier, Alert Engine, Node.js Backend
 - Protocols: HTTPS, Emergency Alerts
- **Database Server:** Data persistence and management
 - Components: PostgreSQL, PostgreSQL Database
 - Protocols: SQL Queries

3.6.3 External Systems

- **Traffic Authority:** Official road data source
 - Components: Official Road Data
 - Connection: Emergency Alerts
- **Data Governance:** Data policy and compliance
 - Components: Research Policy, Anonymized Reports
 - Purpose: Privacy and regulatory compliance

Deployment Considerations:

- **Scalability:** Horizontal scaling capabilities for increased user load
- **Security:** Encrypted communication channels and secure data storage
- **Reliability:** Redundancy and failover mechanisms
- **Performance:** Load balancing and caching strategies



4. Design Analysis

4.1 Structural Analysis

The structural diagrams (Context, Class, and Deployment) provide a comprehensive view of the system's architecture:

- **Context Diagram** effectively identifies all external entities and their interactions with the system, ensuring complete boundary definition
- **Class Diagram** demonstrates a well-structured object-oriented design with clear relationships and responsibilities
- **Deployment Diagram** shows a scalable architecture with proper separation of concerns across different hardware nodes

4.2 Behavioral Analysis

The behavioral diagrams (Use Case, Dataflow, and Sequence) capture the dynamic aspects of the system:

- **Use Case Diagram** comprehensively covers all functional requirements from user and administrator perspectives
- **Dataflow Diagram** illustrates efficient data processing workflows with proper data transformation and storage
- **Sequence Diagrams** demonstrate detailed interaction patterns for key system scenarios, ensuring proper message flow and timing

4.3 Design Strengths

1. **Comprehensive Coverage:** All required UML diagrams provide complete system representation
2. **Clear Separation of Concerns:** Distinct responsibilities across different system components
3. **Scalability:** Architecture supports future expansion and increased user load
4. **User-Centric Design:** Focus on user experience and safety requirements
5. **Real-Time Capabilities:** Effective handling of time-sensitive data and notifications

4.4 Implementation Considerations

1. **Technology Stack:** Modern mobile development frameworks and cloud infrastructure
2. **Data Privacy:** Compliance with data protection regulations and user consent management
3. **Performance Optimization:** Efficient algorithms for real-time processing and route calculation
4. **Offline Functionality:** Robust caching mechanisms for areas with poor connectivity
5. **Testing Strategy:** Comprehensive testing across all diagram scenarios and use cases

5. Conclusion

This system modeling and design document presents a comprehensive UML-based analysis of the Road Sign and Road State Mobile Notification Application. The six required diagrams—Context, Dataflow, Use Case, Sequence, Class, and Deployment diagrams—collectively provide a thorough representation of the system's structure, behavior, and deployment architecture.

5.1 Key Achievements

The comprehensive use of structural, behavioral, and interaction diagrams has been crucial in:

- **Defining System Boundaries:** Clear identification of external entities and system interfaces
- **Modeling Data Flow:** Efficient data processing and transformation workflows
- **Capturing User Requirements:** Complete functional requirement representation through use cases
- **Designing System Structure:** Well-organized object-oriented architecture with clear relationships
- **Planning Deployment:** Scalable physical architecture supporting system requirements
- **Detailing Interactions:** Comprehensive sequence modeling for critical system scenarios

5.2 Design Quality

The resulting design demonstrates:

- **Completeness:** All functional and non-functional requirements addressed
- **Consistency:** Coherent design across all diagram types
- **Scalability:** Architecture capable of handling growth in users and data
- **Maintainability:** Clear structure facilitating future modifications
- **User Safety:** Focus on road safety and minimal driver distraction

5.3 Next Steps

This design documentation provides the foundation for:

1. **Development Phase:** Implementation guidance for development teams
2. **Testing Strategy:** Test case development based on sequence diagrams and use cases
3. **User Interface Design:** Screen design based on use case requirements
4. **Database Design:** Physical database implementation from class relationships
5. **Deployment Planning:** Infrastructure setup based on deployment specifications

The holistic approach to system design ensures that the final product will meet user requirements, adhere to software engineering best practices, and be readily maintainable and scalable over time. This comprehensive modeling serves as the blueprint for successful system implementation and deployment.